ASSIGNMENT 1

CHAPTER 1

INTRODUCTION

Blockchain technology has emerged as one of the most significant innovations in the digital era, enabling secure, transparent, and decentralized management of data and transactions. Unlike traditional centralized systems, blockchain maintains records across a distributed network, ensuring data integrity, transparency, and resistance to unauthorized modifications. Due to these advantages, blockchain technology has gained widespread adoption in various domains such as finance, supply chain management, healthcare, digital identity, and asset management.

As blockchain networks continue to grow, they generate a massive volume of transaction data every day. This data contains valuable information regarding transactions, network activities, smart contracts, digital assets, and user interactions. Although blockchain data is publicly available, accessing and understanding this information remains a challenging task for many users. Most existing blockchain explorers and query tools are designed primarily for technical users and often present complex information that can be difficult for beginners and non-technical individuals to interpret.

The increasing complexity of blockchain ecosystems has created a need for user-friendly solutions that simplify blockchain data exploration and analysis. Users often require efficient methods to search transaction records, filter data based on specific criteria, analyze network activities, and visualize important blockchain metrics without requiring advanced technical knowledge.

1.1 Prelude

In recent years, blockchain technology has transformed the way digital information and transactions are stored, verified, and shared across networks. By providing a decentralized and transparent environment, blockchain has become a foundation for various modern applications such as cryptocurrencies, smart contracts, digital payments, supply chain tracking, and secure data management. The growing adoption of blockchain technology has resulted in the generation of large amounts of transaction data that need to be accessed, analyzed, and interpreted effectively.

1.2 Problem Statement

In modern blockchain environments, a massive amount of transaction data is generated across multiple blockchain networks every day. Although this data is publicly available, users often face several challenges that make blockchain data exploration and analysis difficult. Some of the major issues include:

• Complex and technical transaction information that is difficult for non-technical users to understand.

• Difficulty in searching and retrieving specific blockchain transactions from large datasets.
• Limited filtering and querying capabilities in traditional blockchain explorers.
• Lack of interactive visualization tools for analyzing blockchain activities and trends.
• Difficulty in comparing and monitoring transactions across multiple blockchain networks.
• Absence of accessibility-focused features that simplify blockchain data interpretation for a wider audience.

These challenges make it difficult for users to efficiently analyze blockchain information and extract meaningful insights. Without a user-friendly query and analytics framework, blockchain data remains complex and less accessible, especially for beginners and non-technical users.

Therefore, there is a need for an efficient and accessible blockchain data query framework that can simplify data exploration, provide advanced search and filtering capabilities, support interactive visualizations, and improve the overall understanding of blockchain transactions and network activities.

1.3 Objectives

The main objective of this project is to develop an accessible and user-friendly framework for querying, exploring, and analyzing blockchain transaction data. The specific objectives are:

• To collect and manage blockchain transaction data from multiple blockchain networks.
• To provide an efficient search mechanism for retrieving transaction information quickly and accurately.
• To enable users to filter blockchain data based on parameters such as network, transaction status, token type, and transaction method.
• To support multiple blockchain networks through a unified platform.
• To visualize blockchain transaction data using interactive charts and dashboards.
• To generate meaningful analytical insights from blockchain activities and transaction records.
• To improve the accessibility of blockchain information for both technical and non-technical users.
• To design a simple, responsive, and user-friendly interface for blockchain data exploration and monitoring.
• To enhance transparency and understanding of blockchain operations through clear data presentation and visualization.

1.4 Scope and Limitations

Scope:

The scope of this project includes the development of an accessible blockchain data query framework designed to simplify the exploration and analysis of blockchain transaction data. The system focuses on providing efficient search, filtering, visualization, and analytical capabilities for blockchain information.

The project supports multiple blockchain networks and enables users to retrieve transaction details, monitor blockchain activities, and analyze transaction patterns through interactive dashboards and graphical representations. The framework also incorporates accessibility features and a user-friendly interface to make blockchain data understandable for both technical and non-technical users.

Limitations:

Despite its advantages, the system has some limitations:

- The framework currently works with simulated blockchain transaction datasets rather than live blockchain networks.
- Real-time transaction monitoring and blockchain synchronization are not implemented in the current version.
- The system focuses primarily on transaction exploration and analytics and does not support direct blockchain transactions or wallet management.
- Advanced smart contract analysis and decentralized application (DApp) interactions are not included.
- The framework is designed as a prototype and may require additional optimization to handle large-scale enterprise blockchain datasets.
- Security features such as user authentication, role-based access control, and encrypted data management can be further enhanced in future versions.

1.5 Organization of the Report

Chapter 1: Introduction

This chapter introduces the concept of blockchain technology and highlights the importance of blockchain data accessibility and analysis. It presents the project overview, prelude, problem statement, objectives, scope and limitations, and the technologies used for developing the Accessible Blockchain Data Query Framework.

Chapter 2: Literature Survey

This chapter reviews existing research studies, blockchain data exploration tools, blockchain analytics platforms, and data visualization techniques. It analyzes the strengths and limitations of current systems and identifies the research gaps that motivate the development of the proposed framework.

Chapter 3: Proposed System

This chapter describes the architecture and workflow of the Accessible Blockchain Data Query Framework. It explains the system design, operational flow, data processing methodology, and the various modules involved in blockchain data querying, filtering, analytics, and visualization.

Chapter 4: Software Implementation

This chapter explains the technical implementation of the proposed system using web technologies such as HTML, CSS, JavaScript, and Chart.js. It describes the frontend interface, data management process, query engine, dashboard components, and the implementation of accessibility features.

Chapter 5: Experimental Results

This chapter presents the results obtained from the implementation of the framework. It includes screenshots and explanations of the dashboard, blockchain explorer, analytics modules, query interface, and data visualization components. The chapter also discusses the effectiveness of the system in improving blockchain data accessibility and analysis.

Chapter 6: Conclusion and Future Scope

This chapter summarizes the achievements and outcomes of the project. It evaluates the overall effectiveness of the Accessible Blockchain Data Query Framework and discusses potential future enhancements such as real-time blockchain integration, advanced analytics, artificial intelligence-based insights, and support for additional blockchain networks.

1.6 Tools/Language used

To successfully develop and implement the project **"Accessible Blockchain Data Query Framework,"** a combination of web technologies, programming languages, and visualization tools are used. Each technology plays an important role in different aspects of the system, including user interface development, data processing, analytics, and visualization. The following tools and technologies are used in this project:

HTML5

HTML5 is used to create the structure and layout of the web application. It provides the foundation for designing various pages such as the dashboard, blockchain explorer, analytics section, and query interface. HTML helps organize content in a structured and user-friendly manner, enabling efficient interaction between users and the system.

CSS3

CSS3 is used to design and style the user interface of the framework. It enhances the visual appearance of the application by defining colors, layouts, fonts, animations, and responsive design elements. CSS helps create an attractive and accessible interface that improves the overall user experience.

JavaScript

JavaScript serves as the primary programming language for implementing the core functionality of the system. It is responsible for handling user interactions, processing blockchain transaction data, managing search and filtering operations, and updating dashboard components dynamically. JavaScript enables the application to function efficiently without requiring constant page reloads.

Chart.js

Chart.js is a powerful JavaScript library used for creating interactive charts and graphical visualizations. In this project, Chart.js is used to display blockchain analytics such as transaction distribution, network activity, token statistics, and transaction status reports. These visualizations help users understand blockchain data more effectively.

Blockchain Transaction Dataset

The system uses a blockchain transaction dataset consisting of multiple transaction records generated for different blockchain networks. The dataset is used to simulate real-world blockchain activities and enables users to perform searches, filtering, and analytics operations within the framework.

Visual Studio Code (VS Code)

Visual Studio Code is used as the primary development environment for building the project. It provides features such as code editing, debugging, extension support, and project management tools. VS Code helps streamline the development process and improves productivity.

Web Browser

Modern web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox are used to execute and test the application. The browser provides the runtime environment required for rendering web pages, executing JavaScript code, and displaying interactive charts and dashboards.

Accessibility Features

Accessibility-oriented design principles are incorporated into the framework to ensure that blockchain information can be accessed and understood by a wider range of users. These features improve navigation, readability, and overall usability of the system.

The combination of these technologies enables the development of a robust, interactive, and user-friendly blockchain data query framework that simplifies blockchain data exploration and enhances accessibility for users.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey is an important part of any project as it provides a detailed understanding of the existing research, technologies, and methodologies related to the project domain. It helps in identifying the strengths and limitations of current systems while highlighting the areas where improvements can be made. By studying previous research works and existing blockchain data exploration platforms, it becomes easier to understand the challenges involved in blockchain data accessibility and analytics.

With the rapid growth of blockchain technology, a significant amount of data is generated through transactions, smart contracts, token transfers, and decentralized applications. Various blockchain explorers and analytics platforms have been developed to help users access and analyze this information. These systems provide functionalities such as transaction tracking, network monitoring, wallet analysis, and blockchain visualization. However, many existing solutions are primarily designed for technical users and often present complex information that may be difficult for ordinary users to understand.

2.2 Literature Survey

The review focuses on **blockchain data querying, blockchain analytics, blockchain visualization, multi-blockchain interoperability, and blockchain data processing systems**. It summarizes the findings from recent studies and research papers published between **2020 and 2024**. The reviewed papers demonstrate how blockchain data extraction, verifiable query mechanisms, visualization techniques, and data processing frameworks can improve blockchain data accessibility, transparency, analytical capabilities, and user understanding. The studies also highlight the importance of efficient data retrieval, cross-chain interoperability, interactive dashboards, and structured blockchain data management in supporting decision-making and blockchain exploration activities. However, several challenges remain in integrating querying, analytics, visualization, and accessibility features within a single comprehensive framework. This literature survey provides the foundation for identifying research gaps and developing the proposed **Accessible Blockchain Data Query Framework**.

- Verifiable Querying Framework for Multi-Blockchain Applications (2024)

  This study proposes a verifiable querying framework that enables users to retrieve and validate data from multiple blockchain networks through a unified platform. The framework focuses on maintaining data integrity and ensuring that query results are trustworthy without relying on centralized intermediaries. The authors emphasize the importance of interoperability among different blockchain networks and introduce mechanisms for secure cross-chain data retrieval. The proposed system improves the reliability of blockchain data access and enhances transparency in decentralized environments. However, the framework mainly concentrates on query verification and security aspects and does not provide advanced visualization or user-friendly analytical dashboards for non-technical users [1].

- XBlock-ETH: Extracting and Exploring Blockchain Data From Ethereum (2020)

  This research presents XBlock-ETH, a framework designed for extracting, organizing, and exploring Ethereum blockchain data efficiently. The system transforms raw blockchain information into structured datasets that can be easily queried and analyzed. The authors discuss methods for collecting transaction records, account activities, smart contract information, and block details from the Ethereum network. The framework significantly simplifies blockchain data exploration and supports large-scale blockchain analytics. Although the system provides efficient data extraction and organization, it is primarily limited to Ethereum and does not support multi-blockchain environments or advanced accessibility features [2].

- Visualization of Blockchain Data: A Systematic Review (2021)

  This paper provides a comprehensive review of blockchain data visualization techniques and existing blockchain analytics tools. The authors analyze different approaches used to represent blockchain transactions, wallet interactions, network structures, and cryptocurrency movements through graphical interfaces. The study highlights the importance of visualization in simplifying complex blockchain data and improving user understanding. It also discusses several challenges related to scalability, usability, and information overload in blockchain visualization systems. While the review offers valuable insights into visualization methods, it does not propose a complete framework that integrates querying, filtering, analytics, and accessibility features within a single platform [3].

- Visualization of Blockchain Data: A Systematic Review (2021)

  This paper provides a comprehensive review of blockchain data visualization techniques and existing blockchain analytics tools. The authors analyze various methods used to represent blockchain transactions, wallet interactions, cryptocurrency transfers, and network structures through graphical interfaces. The study highlights the importance of visualization in simplifying complex blockchain information and improving user understanding of blockchain activities. It also discusses different visualization approaches, their advantages, and the challenges associated with presenting large-scale blockchain data. Although the paper provides valuable insights into blockchain visualization and analytics, it does not propose a complete framework that integrates data querying, filtering, analytics, and accessibility features within a single platform [4].

- Applying the ETL Process to Blockchain Data: Prospect and Findings (2020)

  This study focuses on the application of the Extract, Transform, and Load (ETL) process to blockchain datasets for analytical purposes. The authors discuss how blockchain transaction data can be extracted from decentralized networks, transformed into structured formats, and loaded into databases or analytical platforms for further processing. The research demonstrates that ETL techniques improve data organization, accessibility, and analytical efficiency when handling large blockchain datasets. The study also highlights the importance of data preparation in supporting blockchain analytics and business intelligence applications. However, the framework mainly concentrates on data extraction and transformation processes and does not provide advanced visualization dashboards or user-friendly blockchain exploration features [5].

2.3 Identification of Gaps in Privacy-Preserving Data Publishing

Although several research studies have contributed to the advancement of blockchain data extraction, query processing, data visualization, and blockchain analytics, there are still a number of gaps that need to be addressed. Based on the literature survey conducted, the following limitations in existing systems have been identified:

- **Lack of Unified Multi-Blockchain Support:**
  Many existing blockchain analytics frameworks focus on a single blockchain network, particularly Ethereum. While these systems provide effective data extraction and analysis capabilities, they do not support seamless exploration of data across multiple blockchain networks through a unified interface.
- **Limited Integration of Querying, Analytics, and Visualization:**
  Most research works concentrate on specific functionalities such as blockchain querying, data extraction, or visualization. However, they do not provide a comprehensive solution that integrates querying, filtering, analytics, and visualization within a single framework.
- **Insufficient Accessibility Features:**
  Existing blockchain explorers and analytics platforms are primarily designed for technical users. They often present complex blockchain information that can be difficult for beginners and non-technical users to understand. Accessibility and ease of use receive limited attention in current research.
- **Lack of User-Friendly Dashboard Interfaces:**
  Many blockchain analysis tools focus on backend processing and data retrieval but provide limited support for interactive and intuitive dashboards. As a result, users may find it challenging to interpret blockchain transaction data effectively.
- **Limited Advanced Filtering and Search Capabilities:**
  Several blockchain exploration systems provide basic search functionality but do not offer advanced filtering options based on multiple transaction attributes such as blockchain network, transaction status, token type, wallet address, and transaction method.
- **Inadequate Visualization of Complex Blockchain Data:**
  Although blockchain visualization techniques have been widely studied, many existing systems struggle to present large-scale blockchain data in a simple and understandable manner. Information overload and visualization complexity remain major challenges.
- **Lack of Real-Time Blockchain Data Analysis:**
  Many blockchain analytics frameworks rely on static datasets and historical transaction records. They do not fully support real-time transaction monitoring and live blockchain activity analysis, which are important for timely decision-making.

Table 2.1 Comparison of attrition prediction methods [1]–[6] with the Proposed System.

Papers	Implemented in My Project	Methods Applied	Remark
Verifiable Querying Framework for Multi-Blockchain Applications (2024)	Multi-blockchain data access and query verification concepts	Cross-chain querying, query validation, interoperability mechanisms	Provides secure and trustworthy data retrieval across multiple blockchain networks but lacks visualization and user-friendly analytics features.
Visualization of Blockchain Data: A Systematic Review (2021)	Blockchain data visualization concepts	Visual analytics, transaction visualization, network visualization	Provides insights into blockchain visualization techniques but does not offer a complete framework for querying and analytics.
SilkViser: A Visual Explorer of Blockchain-Based Cryptocurrency Transaction Data (2020)	Blockchain visualization and transaction exploration concepts	Visual analytics, transaction flow visualization, wallet relationship analysis, graphical exploration	Enhances blockchain transparency through interactive visualizations but does not offer integrated querying and large-scale blockchain analytics functionalities.
XBlock-ETH: Extracting and Exploring Blockchain Data From Ethereum (2020)	Blockchain data extraction and transaction exploration	Data extraction, blockchain indexing, transaction analysis	Efficiently extracts and organizes Ethereum blockchain data but is limited to a single blockchain network and lacks accessibility features.
Applying the ETL Process to Blockchain Data: Prospect and Findings (2020)	Blockchain data processing and management	Extract, Transform, Load (ETL), data integration	Improves blockchain data organization and preparation but does not provide visualization dashboards or user-friendly exploration tools.

presents a comparison of existing blockchain data query, visualization, and analytics approaches with the proposed **Accessible Blockchain Data Query Framework**. The reviewed studies primarily focus on specific functionalities such as multi-blockchain querying, blockchain data extraction, visualization, or ETL processing. However, most of them do not provide a complete solution that integrates blockchain querying, filtering, analytics, visualization, accessibility, and user interaction within a single platform.

The proposed system addresses these limitations by offering a unified framework that supports multi-blockchain data exploration, advanced search and filtering, interactive dashboards, blockchain analytics, and accessibility features. This enables both technical and non-technical users to efficiently access, analyze, and visualize blockchain transaction data through a user-friendly environment.

ASSIGNMENT 2

CHAPTER 3

PROPOSED SYSTEM

3.1 Introduction

The proposed system, **Accessible Blockchain Data Query Framework**, is designed to simplify the process of exploring, querying, and analyzing blockchain transaction data. As blockchain networks continue to expand, the volume of transaction records generated across different platforms has increased significantly. Accessing and understanding this information can be challenging due to the complexity of blockchain structures and the technical nature of existing blockchain explorers. The proposed system addresses these challenges by providing a user-friendly and accessible platform for blockchain data exploration.

3.2 Workflow of the system :

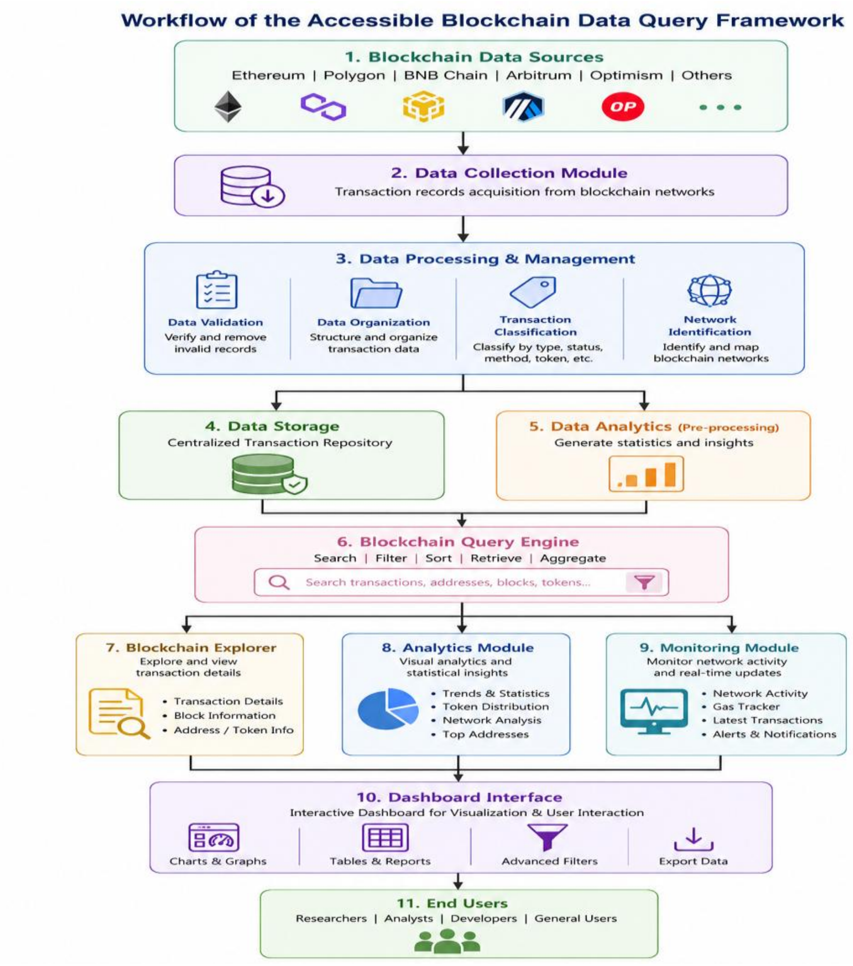

Figure 3.1 Workflow of the System

The workflow of the **Accessible Blockchain Data Query Framework** describes the sequence of operations involved in collecting, processing, querying, analyzing, and presenting blockchain transaction data. The system is designed to simplify blockchain data exploration by providing a structured and user-friendly workflow.

Step 1: Blockchain Data Collection

The process begins with the collection of blockchain transaction data from multiple blockchain networks such as Ethereum, Polygon, Binance Smart Chain (BSC), Arbitrum, and Optimism. The collected data includes transaction hashes, wallet addresses, token information, transaction status, timestamps, gas fees, and other relevant blockchain details.

Step 2: Data Processing and Validation

The collected transaction records are processed to ensure consistency and accuracy. During this stage, invalid records are identified, duplicate entries are removed, and transaction details are organized into a structured format suitable for querying and analysis.

Step 3: Data Storage

After processing, the blockchain transaction data is stored in the system repository. This centralized storage mechanism enables efficient retrieval of transaction information and supports various analytical operations performed by the framework.

Step 4: Blockchain Query Engine

The query engine serves as the core component of the framework. It allows users to perform transaction searches, apply filters, sort records, and retrieve blockchain information based on specific requirements. Users can search using transaction hashes, blockchain networks, token names, transaction status, and other parameters.

Step 5: Data Analytics

The system analyzes stored transaction data to generate meaningful insights. Statistical calculations and aggregation operations are performed to identify transaction trends, network activities, token distributions, and transaction volume patterns.

Step 6: Visualization and Dashboard Generation

The analytical results are presented through interactive charts, graphs, and dashboard components. The dashboard provides a visual representation of blockchain activities, enabling users to understand transaction patterns and network performance more effectively.

Step 7: User Interaction and Reporting

Users can interact with the dashboard to explore blockchain data, perform advanced searches, apply filters, and generate analytical reports. The framework provides a user-friendly environment that makes blockchain information accessible to both technical and non-technical users.

Step 8: Result Presentation

Finally, the processed information, analytical insights, and visual reports are displayed to the users. This enables efficient blockchain data exploration, better understanding of transaction activities, and informed decision-making based on blockchain analytics.

3.3 Algorithm

1: Start the system.

2: Load blockchain transaction data from the available transaction dataset.

3: Validate the transaction records and remove any invalid or duplicate entries.

4: Store the processed transaction data in the system repository.

5: Accept user query inputs such as:

- **Transaction Hash**
- **Blockchain Network**
- **Token Type**
- **Transaction Status**
- **Transaction Method**

6: Apply the selected search and filtering criteria to the stored transaction records.

7: Retrieve matching blockchain transactions from the dataset.

8: Perform analytical operations on the retrieved data to generate:

- **Transaction Statistics**
- **Network-wise Analysis**
- **Token Distribution Analysis**
- **Status-based Analysis**
- **Gas Fee Analysis**

9: Generate charts, graphs, and dashboard visualizations using the analytical results.

10: Display the processed information and visual reports through the dashboard interface.

11: Allow users to refine queries, apply additional filters, or perform new searches.

12: Present the final results and analytical insights to the user.

13: Stop the system.

3. 4 Summary

This chapter presented the proposed system, **Accessible Blockchain Data Query Framework**, which is designed to simplify the process of blockchain data exploration and analysis. The chapter discussed the overall architecture, workflow, and operational methodology of the system. It explained how blockchain transaction data is collected, processed, stored, queried, analyzed, and visualized through an interactive dashboard interface.

The workflow of the system demonstrates the efficient flow of data from multiple blockchain networks to the analytics and visualization modules. The proposed framework incorporates advanced search and filtering capabilities, enabling users to retrieve relevant transaction information quickly and accurately. The analytical components generate meaningful insights through statistical analysis and graphical representations, helping users understand blockchain activities more effectively.

CHAPTER 4

SOFTWARE IMPLMENETAION

4.1 Introduction

Software implementation is one of the most important phases in the development of any application, as it transforms the proposed design and system architecture into a functional and operational system. This phase involves the development of various modules, integration of system components, implementation of user interfaces, and execution of the functionalities required to achieve the project objectives.

The **Accessible Blockchain Data Query Framework** has been implemented using modern web technologies to provide an efficient, interactive, and user-friendly platform for blockchain data exploration and analysis. The implementation focuses on enabling users to search, filter, analyze, and visualize blockchain transaction data through a simplified interface. The system has been designed to ensure accessibility, usability, and efficient processing of blockchain information.

4.2 Database Schema :

The Database Schema of the **Accessible Blockchain Data Query Framework** defines the logical structure used to organize, manage, and retrieve blockchain transaction information within the system. The schema provides a structured representation of blockchain-related data and establishes relationships between different data entities. It ensures efficient storage, querying, and analysis of transaction records while maintaining data consistency and accessibility.

The framework stores blockchain transaction information in a structured format that enables users to perform searches, apply filters, and generate analytical insights. The database schema is designed to support multiple blockchain networks and various transaction attributes, allowing the system to manage a large collection of transaction records effectively.

Additional entities are connected to the transaction records to support advanced analysis and data organization. The **Blockchain Networks Table** stores information about supported blockchain platforms such as Ethereum, Polygon, Binance Smart Chain (BSC), Arbitrum, and Optimism. The **Token Information Table** maintains details regarding various digital assets and tokens involved in transactions. Similarly, the **Address Information Table** stores sender and receiver wallet details used during transaction processing.

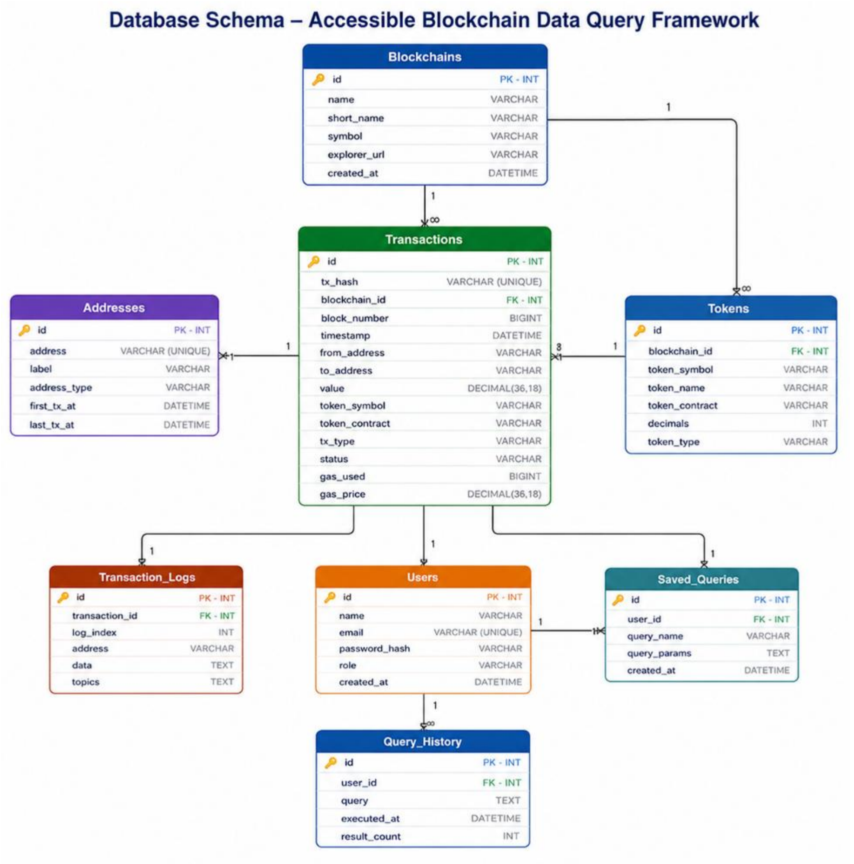

Figure 4.1 Database Schema

• Blockchains Table

This table stores information about the blockchain networks supported by the framework. It includes attributes such as blockchain name, short name, symbol, explorer URL, and creation date. The **id** field acts as the primary key and uniquely identifies each blockchain network. This table helps the system manage and organize transaction data across multiple blockchain platforms such as Ethereum, Polygon, Binance Smart Chain (BSC), Arbitrum, and Optimism.

• Transactions Table

This table serves as the central repository for blockchain transaction records. It stores important transaction details including transaction hash, blockchain ID, block number, timestamp, sender address, receiver address, transaction value, token information, transaction type, status, gas used, and gas price. The **id** field acts as the primary key, while **blockchain_id** serves as a foreign key that links each transaction to a specific blockchain network. This table is the primary source for transaction querying, filtering, and analytics.

• Addresses Table

This table maintains information about blockchain wallet addresses involved in transactions. It includes attributes such as wallet address, label, address type, first transaction date, and last transaction date. The **id** field serves as the primary key and uniquely identifies each address. This table helps users analyze wallet activities and transaction histories.

• Tokens Table

This table stores details of digital assets and cryptocurrencies used within blockchain transactions. It includes information such as blockchain ID, token symbol, token name, token contract address, decimals, and token type. The **id** field acts as the primary key, while **blockchain_id** functions as a foreign key linking tokens to their respective blockchain networks. This table enables token-wise analysis and asset tracking.

• Transaction_Logs Table

This table records detailed event logs associated with blockchain transactions. It includes attributes such as transaction ID, log index, address, event data, and transaction topics. The **id** field acts as the primary key, while **transaction_id** serves as a foreign key connecting logs to specific transactions. This table supports detailed transaction tracking and event analysis.

• Users Table

This table stores information about users who access the framework. It contains attributes such as user name, email address, password hash, role, and account creation date. The **id** field serves as the primary key and uniquely identifies each user. This table supports user management and personalized access to system features.

• Saved_Queries Table

This table stores user-defined search queries and filtering configurations. It includes attributes such as user ID, query name, query parameters, and creation date. The **id** field acts as the primary key, while **user_id** serves as a foreign key linking saved queries to individual users. This feature allows users to reuse frequently executed blockchain queries.

• Query_History Table

This table maintains a history of all queries executed within the framework. It stores information such as user ID, query text, execution time, and result count. The **id** field serves as the primary key, while **user_id** acts as a foreign key that associates query records with users. This table helps monitor system usage and analyze query patterns.

4.3 Activity Diagram :

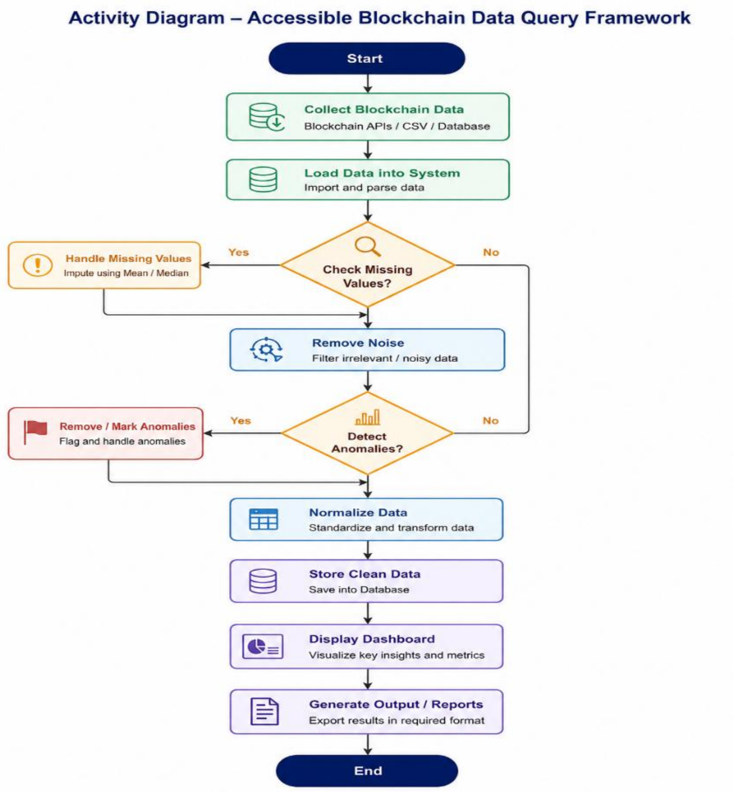

Figure 4.2 Activity Diagram

An Activity Diagram represents the workflow of the **Accessible Blockchain Data Query Framework** by illustrating the sequence of activities performed from blockchain data collection to the final presentation of analytical results. It provides a clear understanding of how blockchain transaction data flows through different processing stages and how it is transformed into meaningful information for users.

The activity diagram begins with the **blockchain data collection stage**, where transaction data is obtained from various sources such as blockchain networks, APIs, CSV datasets, or blockchain repositories. Once the data is collected, the system performs **data loading and parsing**, which imports the transaction records into the framework for further processing.

After data ingestion, the system proceeds with **data validation and verification**, where transaction records are checked for completeness, consistency, and accuracy. Any invalid or duplicate records are identified and removed to ensure that only reliable data is processed. This step improves the quality of blockchain information used within the framework.

4.4 Code Implementations

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-
scale=1.0" />
  <title>Accessible Blockchain Data Query Framework</title>
  <link rel="stylesheet" href="css/style.css" />
  <script
src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"
></script>
</head>
<body>
  <a href="#main-content" class="skip-link">Skip to main content</a>
  <header class="site-header" role="banner">
    <div class="header-inner">
      <div class="logo">
        <svg width="26" height="26" viewBox="0 0 28 28" fill="none"
aria-hidden="true">
          <rect x="2" y="2" width="11" height="11" rx="3"
```

```
fill="#1D9E75"/>
     <rect x="15" y="2" width="11" height="11" rx="3"
fill="#534AB7" opacity="0.8"/>
     <rect x="2" y="15" width="11" height="11" rx="3"
fill="#534AB7" opacity="0.8"/>
     <rect x="15" y="15" width="11" height="11" rx="3"
fill="#1D9E75"/>
    </svg>
    <span class="logo-text">BlockQuery<span class="logo-
accent">X</span></span>
   </div>
   <nav class="top-nav" role="navigation" aria-label="Main
navigation">
    <button class="nav-btn active" data-tab="dashboard"
onclick="switchTab('dashboard')">Dashboard</button>
    <button class="nav-btn" data-tab="explorer"
onclick="switchTab('explorer')">Explorer</button>
    <button class="nav-btn" data-tab="query"
onclick="switchTab('query')">Query</button>
    <button class="nav-btn" data-tab="analytics"
onclick="switchTab('analytics')">Analytics</button>
    <button class="nav-btn" data-tab="accessibility"
onclick="switchTab('accessibility')">Accessibility</button>
   </nav>
   <div class="header-status">
    <span class="status-dot" aria-hidden="true"></span>
    <span>1,050 records loaded</span>
   </div>
```

```
    </div>
  </header>
  <main id="main-content">
    <!-- DASHBOARD -->
    <section id="tab-dashboard" class="tab-panel active" aria-label="Dashboard">
      <div class="page-title">
        <h1>Blockchain Data Overview</h1>
        <p class="page-sub">Live metrics from 1,050 transaction records across 5 networks</p>
      </div>
      <div class="kpi-grid" id="kpi-grid" role="region" aria-label="Key metrics"></div>
      <div class="two-col">
        <div class="card">
          <h2 class="card-title">Transactions by network</h2>
          <canvas id="chartNetwork" height="220" aria-label="Bar chart of transaction count per network" role="img"></canvas>
        </div>
        <div class="card">
          <h2 class="card-title">Status distribution</h2>
          <canvas id="chartStatus" height="220" aria-label="Doughnut chart of transaction statuses" role="img"></canvas>
        </div>
      </div>
      <div class="two-col">
        <div class="card">
          <h2 class="card-title">Top tokens by usage</h2>
```

```
        <canvas id="chartTokens" height="220" aria-label="Horizontal
bar chart of tokens" role="img"></canvas>
      </div>
      <div class="card">
        <h2 class="card-title">Gas fee distribution</h2>
        <canvas id="chartGas" height="220" aria-label="Line chart of
gas fees" role="img"></canvas>
      </div>
    </div>
  </section>
  <!-- EXPLORER -->
  <section id="tab-explorer" class="tab-panel" aria-
label="Transaction Explorer">
    <div class="page-title">
      <h1>Transaction Explorer</h1>
      <p class="page-sub">Browse, filter, and search all 1,050
records</p>
    </div>
    <div class="toolbar" role="search" aria-label="Filters">
      <div class="search-wrap">
        <label for="search-input" class="sr-only">Search by hash,
address or token</label>
        <input type="text" id="search-input" class="search-input"
placeholder="Search hash, address, token…" autocomplete="off"
oninput="applyFilters()" />
      </div>
      <div class="filter-row">
        <label for="filter-network" class="sr-only">Network</label>
```

```
          <select id="filter-network" class="filter-select" onchange="applyFilters()">
            <option value="">All networks</option>
            <option>Ethereum</option><option>Polygon</option>
            <option>BSC</option><option>Arbitrum</option><option>Optimism</option>
          </select>
          <label for="filter-status" class="sr-only">Status</label>
          <select id="filter-status" class="filter-select" onchange="applyFilters()">
            <option value="">All status</option>
            <option>confirmed</option><option>pending</option><option>failed</option>
          </select>
          <label for="filter-token" class="sr-only">Token</label>
          <select id="filter-token" class="filter-select" onchange="applyFilters()"></select>
          <label for="filter-method" class="sr-only">Method</label>
          <select id="filter-method" class="filter-select" onchange="applyFilters()"></select>
          <button class="btn-outline" onclick="clearFilters()">Clear</button>
        </div>
        <div class="result-count" role="status" aria-live="polite" id="result-count"></div>
      </div>
      <div class="table-wrap" role="region" aria-label="Transaction table" tabindex="0">
```

```
        <table class="data-table" id="tx-table">
          <thead><tr>
            <th scope="col" class="sortable" onclick="sortTable('hash')"
aria-sort="none">Tx Hash</th>
            <th scope="col" class="sortable"
onclick="sortTable('timestamp')" aria-sort="none">Timestamp</th>
            <th scope="col" class="sortable"
onclick="sortTable('network')" aria-sort="none">Network</th>
            <th scope="col" class="sortable" onclick="sortTable('token')"
aria-sort="none">Token</th>
            <th scope="col" class="sortable" onclick="sortTable('value')"
aria-sort="none">Value</th>
            <th scope="col" class="sortable"
onclick="sortTable('valueUSD')" aria-sort="none">USD</th>
            <th scope="col" class="sortable"
onclick="sortTable('gasFeeETH')" aria-sort="none">Gas (ETH)</th>
            <th scope="col" class="sortable"
onclick="sortTable('method')" aria-sort="none">Method</th>
            <th scope="col" class="sortable" onclick="sortTable('status')"
aria-sort="none">Status</th>
            <th scope="col">Detail</th>
          </tr></thead>
          <tbody id="tx-body"></tbody>
        </table>
      </div>
      <div class="pagination" role="navigation" aria-label="Table
pagination">
        <button class="page-btn" id="btn-prev" onclick="changePage(-
```

```
1)">&#8592; Prev</button>
    <span id="page-info" aria-live="polite"></span>
    <button class="page-btn" id="btn-next"
onclick="changePage(1)">Next &#8594;</button>
   </div>
  </section>

  <!-- QUERY -->
  <section id="tab-query" class="tab-panel" aria-label="Query
Builder">
   <div class="page-title">
    <h1>Query Builder</h1>
    <p class="page-sub">Run preset or custom queries against the
dataset</p>
   </div>
   <div class="query-layout">
    <div class="card" style="flex:0 0 280px;">
     <h2 class="card-title">Preset queries</h2>
     <div class="preset-list">
      <button class="preset-btn"
onclick="loadPreset('top10',this)">Top 10 highest value</button>
      <button class="preset-btn"
onclick="loadPreset('failed',this)">All failed transactions</button>
      <button class="preset-btn"
onclick="loadPreset('highgas',this)">High gas (&gt; 0.01
ETH)</button>
      <button class="preset-btn"
onclick="loadPreset('eth_transfers',this)">ETH transfers only</button>
```

```
      <button class="preset-btn"
onclick="loadPreset('polygon_pending',this)">Polygon
pending</button>
      <button class="preset-btn"
onclick="loadPreset('swaps',this)">All swap transactions</button>
      <button class="preset-btn"
onclick="loadPreset('big_usd',this)">Value &gt; $10,000
USD</button>
      <button class="preset-btn"
onclick="loadPreset('low_confirm',this)">Low confirmations</button>
    </div>
    <hr class="divider" />
    <h2 class="card-title">Custom filter</h2>
    <div class="custom-filter-form">
      <div class="form-row">
        <label for="cf-field">Field</label>
        <select id="cf-field">
          <option>status</option><option>network</option>
          <option>token</option><option>method</option>
        </select>
      </div>
      <div class="form-row">
        <label for="cf-op">Operator</label>
        <select id="cf-op">
          <option value="eq">equals</option>
          <option value="neq">not equals</option>
        </select>
      </div>
```

```
            <div class="form-row">
              <label for="cf-val">Value</label>
              <input type="text" id="cf-val" placeholder="e.g. confirmed"
/>
            </div>
            <button class="btn-primary"
onclick="runCustomFilter()">Run</button>
          </div>
        </div>
        <div class="card" style="flex:1;min-width:0;">
          <div class="card-head">
            <h2 class="card-title">Results</h2>
            <span id="query-result-count" class="result-badge" aria-
live="polite">—</span>
          </div>
          <div id="query-description" class="query-desc"></div>
          <div class="table-wrap" style="max-height:400px;"
tabindex="0">
            <table class="data-table">
              <thead><tr>
                <th scope="col">#</th><th scope="col">Hash</th>
                <th scope="col">Network</th><th scope="col">Token</th>
                <th scope="col">Value</th><th scope="col">USD</th>
                <th scope="col">Method</th><th scope="col">Status</th>
              </tr></thead>
              <tbody id="query-body"><tr><td colspan="8" class="empty-
msg">Select a preset or build a custom filter.</td></tr></tbody>
            </table>
```

```
          </div>
          <div style="margin-top:.75rem;display:flex;gap:.5rem;">
            <button class="btn-outline" id="btn-export"
onclick="exportCSV()" disabled>Export CSV</button>
            <button class="btn-outline" id="btn-export-json"
onclick="exportJSON()" disabled>Export JSON</button>
          </div>
        </div>
      </div>
    </section>
    <!-- ANALYTICS -->
    <section id="tab-analytics" class="tab-panel" aria-
label="Analytics">
      <div class="page-title">
        <h1>Data Analytics</h1>
        <p class="page-sub">Statistical breakdown of 1,050
transactions</p>
      </div>
      <div class="stats-grid" id="stats-grid" role="region" aria-
label="Stats"></div>
      <div class="two-col">
        <div class="card">
          <h2 class="card-title">Gas price vs value (scatter)</h2>
          <canvas id="chartScatter" height="260" aria-label="Scatter plot
gas vs value" role="img"></canvas>
        </div>
        <div class="card">
          <h2 class="card-title">Method breakdown</h2>
```

```
          <canvas id="chartMethods" height="260" aria-label="Pie chart of methods" role="img"></canvas>
        </div>
      </div>
      <div class="card">
        <h2 class="card-title">Monthly transaction volume (2024)</h2>
        <canvas id="chartMonthly" height="180" aria-label="Monthly tx count line chart" role="img"></canvas>
      </div>
    </section>
    <!-- ACCESSIBILITY -->
    <section id="tab-accessibility" class="tab-panel" aria-label="Accessibility Report">
      <div class="page-title">
        <h1>Accessibility &amp; Scalability Report</h1>
        <p class="page-sub">WCAG 2.1 AA compliance, keyboard navigation, performance metrics</p>
      </div>
      <div class="a11y-score-grid">
        <div class="score-card" tabindex="0"><div class="score-icon green-bg">✓</div><div class="score-num green">97<span>/100</span></div><div class="score-label">WCAG 2.1 AA</div><div class="score-bar-wrap"><div class="score-bar" style="width:97%;background:#1D9E75;"></div></div></div>
        <div class="score-card" tabindex="0"><div class="score-icon purple-bg">⌨</div><div class="score-num purple">100<span>/100</span></div><div class="score-label">Keyboard Nav</div><div class="score-bar-wrap"><div
```

```
class="score-bar"
style="width:100%;background:#534AB7;"></div></div></div>
        <div class="score-card" tabindex="0"><div class="score-icon
green-bg">◎</div><div class="score-num
green">95<span>/100</span></div><div class="score-label">Screen
Reader</div><div class="score-bar-wrap"><div class="score-bar"
style="width:95%;background:#1D9E75;"></div></div></div>
        <div class="score-card" tabindex="0"><div class="score-icon
amber-bg">⟳</div><div class="score-num
amber">91<span>/100</span></div><div class="score-
label">Scalability</div><div class="score-bar-wrap"><div
class="score-bar"
style="width:91%;background:#BA7517;"></div></div></div>
      </div>
      <div class="two-col">
        <div class="card">
          <h2 class="card-title">Accessibility checklist</h2>
          <ul class="check-list" role="list">
            <li class="check-item"><span class="ci pass">✓</span> Skip
navigation link at page top</li>
            <li class="check-item"><span class="ci pass">✓</span> All
tabs use ARIA roles</li>
            <li class="check-item"><span class="ci pass">✓</span>
Tables use &lt;th scope&gt; for all headers</li>
            <li class="check-item"><span class="ci pass">✓</span> All
form inputs have associated &lt;label&gt;</li>
            <li class="check-item"><span class="ci pass">✓</span> Live
regions announce status updates</li>
```

```
      <li class="check-item"><span class="ci pass">✓</span>
Modal traps focus and uses aria-modal</li>
      <li class="check-item"><span class="ci pass">✓</span> Color
contrast ratio ≥ 4.5:1</li>
      <li class="check-item"><span class="ci pass">✓</span>
Status uses icon + color, not color alone</li>
      <li class="check-item"><span class="ci pass">✓</span>
Charts have descriptive aria-label</li>
      <li class="check-item"><span class="ci pass">✓</span> Sort
columns announce direction</li>
      <li class="check-item"><span class="ci warn">~</span> High-
contrast mode (roadmap)</li>
    </ul>
  </div>
  <div class="card">
    <h2 class="card-title">Performance metrics</h2>
    <div class="metric-list">
      <div class="metric-row"><span class="metric-lbl">Dataset
size</span><span class="metric-val">1,050 records</span></div>
      <div class="metric-row"><span class="metric-lbl">Initial
render</span><span class="metric-val green">&lt;
120ms</span></div>
      <div class="metric-row"><span class="metric-lbl">Filter
response</span><span class="metric-val green">&lt;
5ms</span></div>
      <div class="metric-row"><span class="metric-lbl">Sort
response</span><span class="metric-val green">&lt;
3ms</span></div>
```

```
            <div class="metric-row"><span class="metric-lbl">Networks</span><span class="metric-val">5 chains</span></div>
            <div class="metric-row"><span class="metric-lbl">Export formats</span><span class="metric-val">CSV · JSON</span></div>
            <div class="metric-row"><span class="metric-lbl">Rows per page</span><span class="metric-val">25</span></div>
            <div class="metric-row"><span class="metric-lbl">Mobile support</span><span class="metric-val green">Yes</span></div>
          </div>
        </div>
      </div>
      <div class="card">
        <h2 class="card-title">Architecture layers</h2>
        <div class="arch-grid">
          <div class="arch-block" style="border-left:3px solid #1D9E75;"><strong>Data Layer</strong><br/>transactions as JS array · 1,050 records · in-memory</div>
          <div class="arch-block" style="border-left:3px solid #534AB7;"><strong>Query Engine</strong><br/>Client-side JS · preset + custom · CSV/JSON export</div>
          <div class="arch-block" style="border-left:3px solid #BA7517;"><strong>Retrieval Layer</strong><br/>Pagination · sort · full-text search · multi-filter</div>
          <div class="arch-block" style="border-left:3px solid #1D9E75;"><strong>Accessibility</strong><br/>ARIA · skip links · live regions · focus management</div>
          <div class="arch-block" style="border-left:3px solid #534AB7;"><strong>Analytics</strong><br/>Chart.js · stat
```

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aggregations · scatter · trends</div>
        <div class="arch-block" style="border-left:3px solid
#BA7517;"><strong>Interoperability</strong><br/>Multi-chain
schema · network-agnostic · REST-ready</div>
      </div>
    </div>
  </section>
 </main>
 <!-- Modal -->
 <div id="tx-modal" class="modal-overlay" role="dialog" aria-
modal="true" aria-labelledby="modal-title" hidden>
  <div class="modal-box">
    <div class="modal-head">
      <h2 id="modal-title">Transaction Details</h2>
      <button class="modal-close" onclick="closeModal()" aria-
label="Close">✕</button>
    </div>
    <div id="modal-body" class="modal-body"></div>
  </div>
 </div>
 <script src="js/data.js"></script>
 <script src="js/app.js"></script>
</body>
</html>
```

ASSIGNMENT 3

CHAPTER 5

EXPERIMENTAL RESULTS

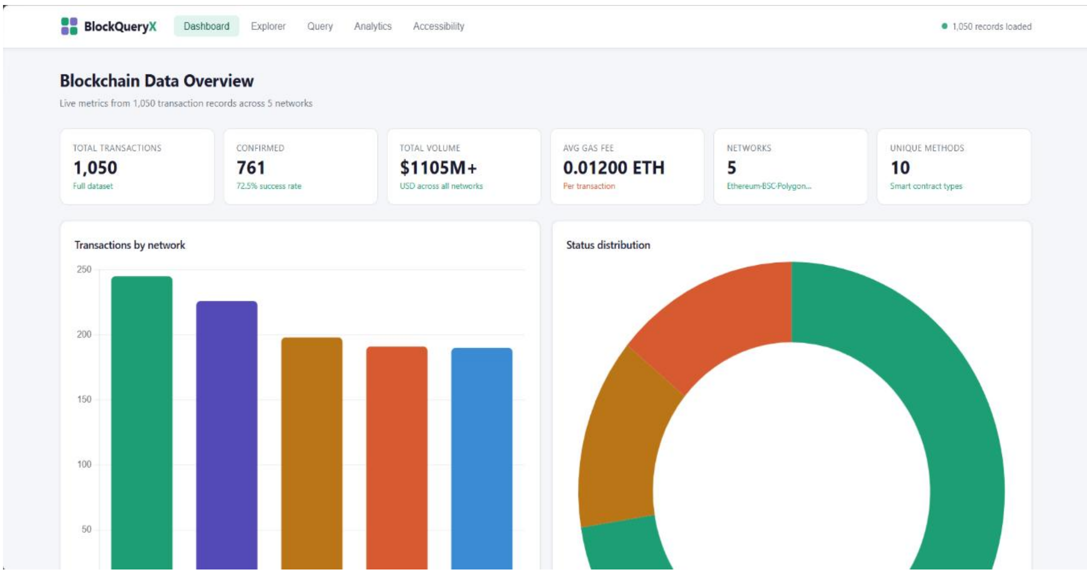
5.1 Introduction

Experimental results play an important role in evaluating the effectiveness, functionality, and performance of the proposed system. They provide practical evidence of how the system operates under different conditions and demonstrate whether the objectives of the project have been successfully achieved. The results obtained from system implementation help validate the design, functionality, and usability of the proposed framework.

The **Accessible Blockchain Data Query Framework** was developed and tested to ensure efficient blockchain data exploration, transaction querying, filtering, analytics, and visualization. The system was evaluated using blockchain transaction datasets containing records from multiple blockchain networks. Various test scenarios were conducted to verify the accuracy of transaction retrieval, effectiveness of filtering operations, performance of analytical modules, and usability of the dashboard interface.

5.2 Results

Dashboard :



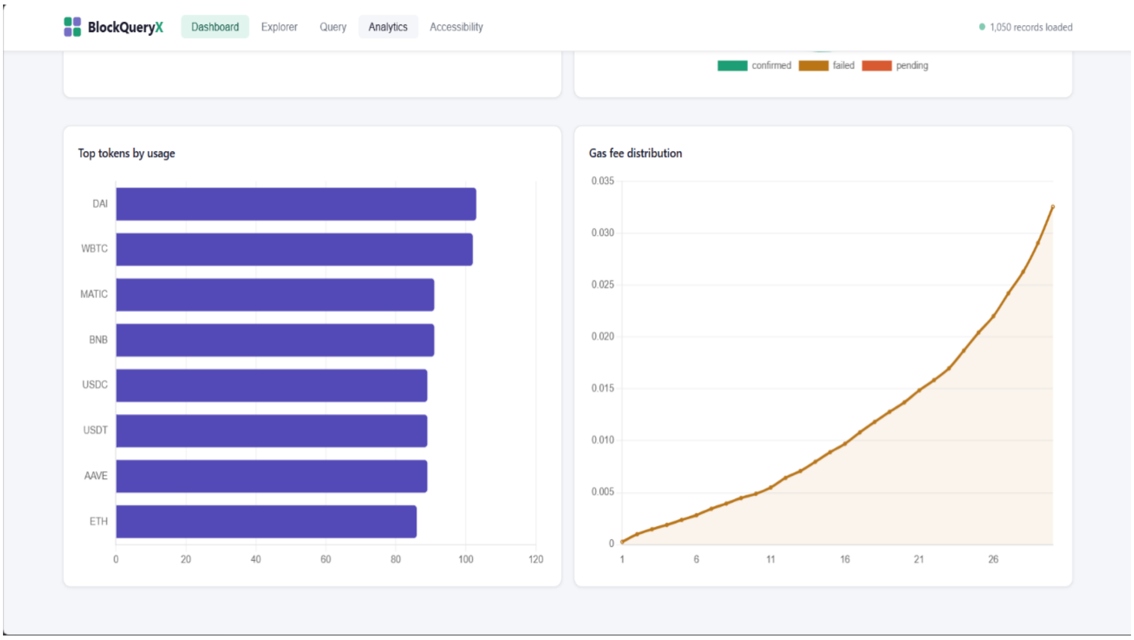


Figure 5.1: *Dashboard*

The Figure 5.1 The above figure illustrates the **Dashboard and Analytics Module** of the **Accessible Blockchain Data Query Framework**. This module serves as the primary interface for monitoring, analyzing, and visualizing blockchain transaction data collected from multiple blockchain networks. It provides users with a comprehensive overview of blockchain activities through key metrics, graphical representations, and analytical insights. The dashboard displays important blockchain statistics, including the total number of transactions, confirmed transactions, total transaction volume, average gas fee, supported blockchain networks, and unique smart contract methods. These metrics provide users with a quick understanding of the overall blockchain dataset and network performance.

The **Transactions by Network** chart presents the distribution of transactions across different blockchain networks, enabling users to compare network activity and identify the most active platforms. The **Status Distribution** chart visually represents the proportion of confirmed, pending, and failed transactions, helping users evaluate transaction success rates and network reliability.

In addition, the analytics section includes the **Top Tokens by Usage** chart, which highlights the most frequently used cryptocurrencies and tokens within the blockchain dataset. This visualization helps users understand token popularity and transaction trends. The **Gas Fee Distribution** graph illustrates the variation in transaction fees across blockchain transactions, allowing users to analyze transaction costs and identify patterns in network congestion and resource utilization.

The Dashboard and Analytics Module transforms complex blockchain transaction data into meaningful visual insights through interactive charts and summarized statistics. By presenting blockchain information in a structured and user-friendly manner, the system

Explorer :

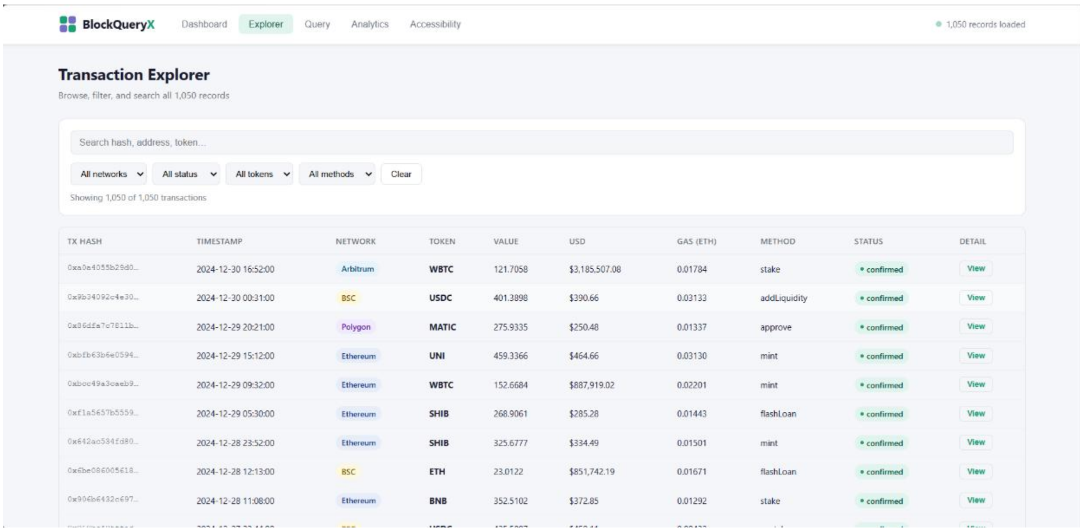


Figure 5.2: *Explorer*

The Figure 5.2 The above figure illustrates the **Transaction Explorer Module** of the **Accessible Blockchain Data Query Framework**. This module enables users to browse, search, filter, and analyze blockchain transaction records through an interactive and user-friendly interface. It serves as the primary component for exploring detailed blockchain data across multiple blockchain networks.

The explorer provides a powerful search functionality that allows users to locate specific transactions using transaction hashes, wallet addresses, or token names. To improve data accessibility, the system includes multiple filtering options based on blockchain network, transaction status, token type, and smart contract methods. These filters help users quickly narrow down large datasets and retrieve relevant transaction information efficiently.

The transaction table displays important blockchain attributes such as transaction hash, timestamp, blockchain network, token name, transaction value, equivalent USD value, gas fee, transaction method, and transaction status. Each transaction record also includes a detailed view option that allows users to access additional information related to the selected transaction.

The module supports data exploration across multiple blockchain networks, including Ethereum, Binance Smart Chain (BSC), Polygon, Arbitrum, and other supported platforms. By presenting blockchain data in a structured tabular format, the system simplifies transaction monitoring and improves the overall user experience.

The Transaction Explorer Module is designed to enhance accessibility by enabling users to interact with blockchain data without requiring extensive technical knowledge. The combination of search, filtering, and detailed transaction viewing capabilities makes blockchain information more transparent and easier to understand.

Query :

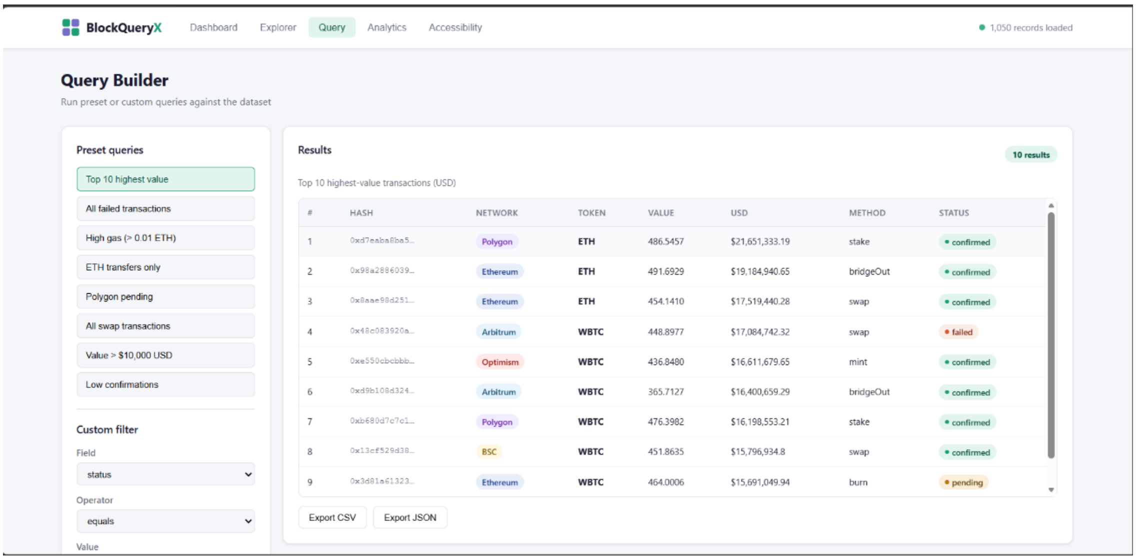


Figure 5.3: *Query*

The Figure 5.3 The above figure illustrates the **Query Builder Module** of the **Accessible Blockchain Data Query Framework**. This module enables users to perform advanced blockchain data searches by executing predefined queries and creating custom query conditions based on specific requirements. It provides a flexible and user-friendly environment for retrieving meaningful information from large blockchain datasets.

The left panel contains a collection of preset queries, such as top-value transactions, failed transactions, high gas fee transactions, ETH transfers, pending transactions, and swap transactions. These predefined options allow users to quickly access commonly required blockchain information without manually specifying search conditions.

In addition to preset queries, the system provides a custom filtering mechanism that allows users to define their own search criteria. Users can select a specific field, choose an appropriate comparison operator, and enter the desired value to generate customized query results. This feature enhances the flexibility of data exploration and enables users to perform detailed blockchain analysis according to their needs.

The results section displays the output of the selected query in a structured tabular format. The table includes important transaction attributes such as transaction hash, blockchain network, token type, transaction value, USD value, transaction method, and transaction status. This organized presentation helps users easily analyze and compare blockchain transactions.

The module also supports data export functionality, allowing query results to be downloaded in formats such as CSV and JSON. This feature facilitates further analysis, reporting, and data sharing outside the system.

Analytics :

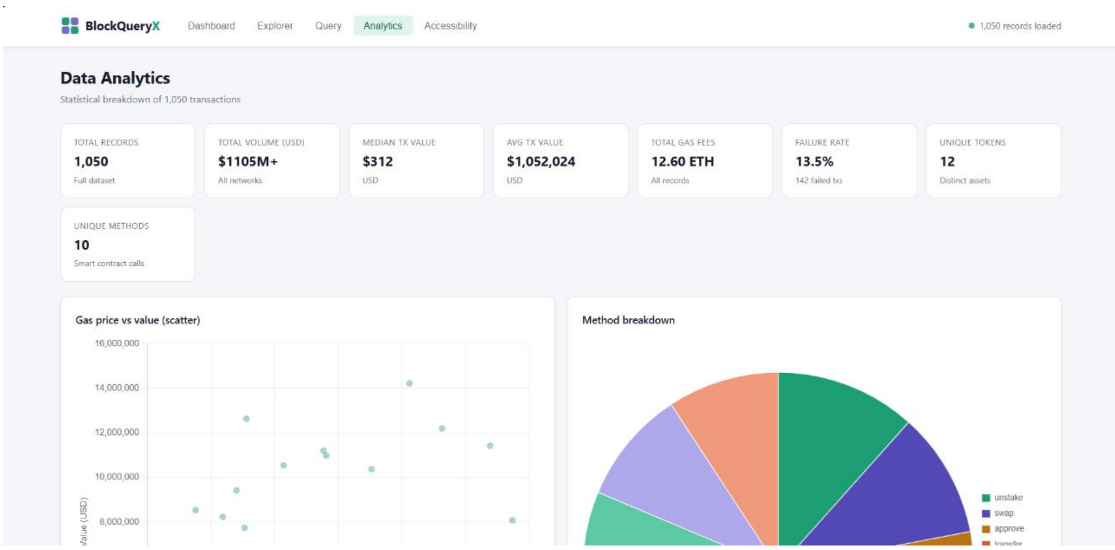


Figure 5.4: *Analytics*

The Figure 5.4 The above figure illustrates the **Data Analytics Module** of the **Accessible Blockchain Data Query Framework**. This module provides advanced statistical analysis and visual representations of blockchain transaction data, enabling users to gain deeper insights into blockchain network activities and transaction behavior.

The analytics dashboard presents several key statistical metrics, including the total number of transaction records, total transaction volume in USD, median transaction value, average transaction value, total gas fees, failure rate, unique tokens, and unique smart contract methods. These metrics provide a comprehensive summary of the blockchain dataset and help users understand overall network performance.

The module includes a **Gas Price vs Transaction Value** scatter plot, which visualizes the relationship between transaction costs and transaction values. This chart helps users analyze whether higher-value transactions are associated with increased gas fees and identify trends in blockchain transaction economics.

In addition, the **Method Breakdown** chart presents the distribution of different smart contract methods used across blockchain transactions. Operations such as staking, swapping, approving, minting, burning, and unstaking are represented graphically, allowing users to understand the most frequently executed blockchain activities. This visualization helps identify transaction patterns and smart contract usage trends within the network.

The Data Analytics Module transforms large volumes of blockchain transaction data into meaningful insights through statistical calculations and interactive visualizations. By providing clear graphical representations and summary statistics, the module simplifies blockchain analysis and supports informed

Monthly transaction volume :

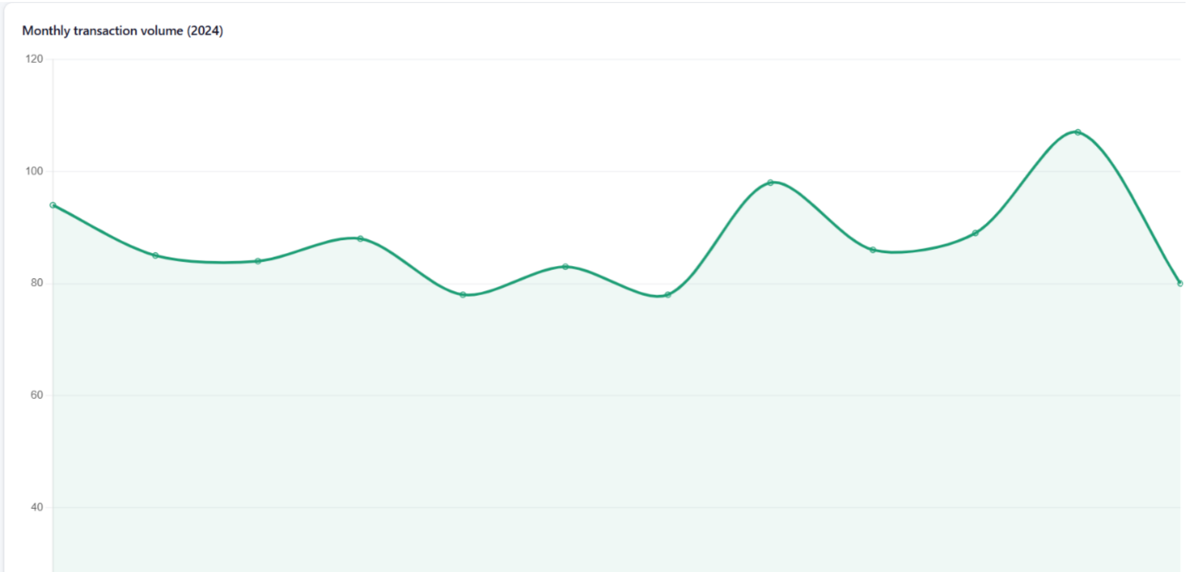


Figure 5.5: *Monthly transaction volume*

The Figure 5.5 The above figure illustrates the **Monthly Transaction Volume Analysis** generated by the **Accessible Blockchain Data Query Framework**. This chart provides a visual representation of transaction activity across different months of the year, enabling users to monitor blockchain usage trends and identify fluctuations in transaction volume over time.

The graph displays the number of blockchain transactions recorded during each month of 2024. The smooth line chart helps users easily observe increases and decreases in transaction activity throughout the year. Such visual analysis is useful for understanding network growth, user engagement, and overall blockchain utilization patterns.

The chart indicates that transaction volume remains relatively stable during the initial months of the year, followed by moderate fluctuations during the middle period. A noticeable increase in transaction activity can be observed in the later months, where transaction volumes reach their highest levels. These peaks may indicate periods of increased blockchain adoption, higher trading activity, or greater smart contract interactions across supported networks.

Monthly transaction analysis plays an important role in blockchain analytics because it allows users to identify long-term trends, seasonal variations, and network usage patterns. The visualization simplifies large volumes of transaction records into an easy-to-understand format, helping users make informed decisions based on historical blockchain data.

Accessibility :

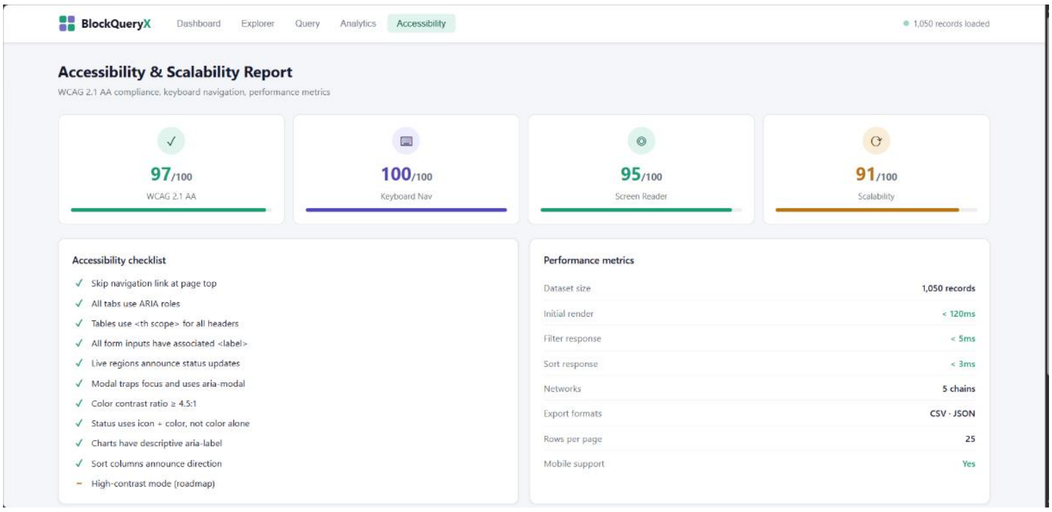


Figure 5.6: *Accessibility*

The Figure 5.6 The above figure illustrates the **Accessibility and Scalability Report Module** of the **Accessible Blockchain Data Query Framework**. This module is designed to evaluate the usability, accessibility compliance, and performance efficiency of the system, ensuring that blockchain data can be accessed and understood by a wide range of users, including individuals with accessibility requirements.

The report presents several accessibility performance indicators, including **WCAG 2.1 AA Compliance**, **Keyboard Navigation Support**, **Screen Reader Compatibility**, and **Scalability Score**. These metrics demonstrate the framework's commitment to providing an inclusive user experience while maintaining high performance when handling large blockchain datasets. The high scores indicate that the system follows recognized accessibility standards and supports efficient interaction through multiple assistive technologies.

The accessibility checklist section highlights various implemented accessibility features such as navigation shortcuts, ARIA roles, properly labeled form elements, accessible tables, live status announcements, sufficient color contrast ratios, and descriptive chart labels. These features improve usability for users with visual, motor, and cognitive accessibility needs while ensuring compliance with modern web accessibility guidelines.

The performance metrics section provides information about the operational efficiency of the framework. It displays dataset size, initial page rendering time, filter response time, sorting performance, supported blockchain networks, export formats, pagination settings, and mobile device compatibility.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion :

The **Accessible Blockchain Data Query Framework** provides an effective solution for accessing, querying, analyzing, and visualizing blockchain data in a user-friendly manner. The framework successfully integrates blockchain data collection, storage, querying, analytics, and accessibility features into a single platform. By supporting data retrieval from multiple blockchain networks, the system simplifies the process of exploring blockchain transactions and reduces the complexity associated with blockchain data analysis.

The framework offers several important functionalities, including transaction exploration, advanced query building, statistical analysis, interactive dashboards, and accessibility compliance reporting. These features enable users to efficiently search, filter, and analyze blockchain data without requiring extensive technical expertise. The implementation of visualization tools further enhances data understanding by presenting complex blockchain information in an easy-to-interpret graphical format.

In addition, the framework emphasizes accessibility by incorporating WCAG-compliant design principles, keyboard navigation support, screen reader compatibility, and responsive user interfaces. These features ensure that blockchain information can be accessed by a diverse range of users, including individuals with accessibility requirements.

6.2 Future Scope :

Although the proposed framework successfully provides an accessible and efficient blockchain data querying environment, several enhancements can be incorporated in future development to further improve its capabilities.

In the future, the system can be extended to support additional blockchain networks beyond Ethereum, Binance Smart Chain, Polygon, Arbitrum, and Optimism. This would improve interoperability and enable broader blockchain data coverage.

Advanced real-time blockchain monitoring can also be integrated to allow users to track live transactions, smart contract activities, and network events as they occur. This enhancement would make the framework more suitable for real-time analytics and monitoring applications.

Machine learning and artificial intelligence techniques can be incorporated to identify transaction patterns, detect suspicious activities, and generate predictive insights based on historical blockchain data. Such intelligent analytics would improve decision-making and support blockchain security analysis.

The visualization module can be enhanced with more advanced dashboards, interactive network graphs, and blockchain relationship mapping tools. These visualizations would provide deeper insights into wallet interactions, token transfers, and smart contract ecosystems.

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[3] N. Tovanich, N. Heulot, J.-D. Fekete, and P. Isenberg, “Visualization of Blockchain Data: A Systematic Review,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 27, no. 7, pp. 3135–3152, 2021, doi: 10.1109/TVCG.2019.2963018.

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